



Review Test Submission: Assignment Week 11

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Course EBN 122 S2 2017

Test Assignment Week 11

Started 10/16/17 8:48 AM

Submitted 10/23/17 5:12 PM

Due Date 10/23/17 11:59 PM

Status Completed

Attempt Score 11.7 out of 20 points

Time Elapsed 176 hours, 24 minutes

Instructions Read over the document *ClickUP Assignments Instructions*.

The deadline is the time by which the assignment must be SUBMITTED (not just started)

Before sending an enquiring regarding ClickUP question to any-one please verify that your problem is valid ... i.e. if you suspect incorrect marking first check with a fellow student or two.

Please direct any enquiries regarding ClickUP assignments to EBN2017.ClickUptest@gmail.com

Save the answer to each question as you complete it.

At the end of the assignment click Save and Submit.

Give your final answer correct to 3 significant digits.

Make sure you use the FULL STOP for your decimal place marker.

You must indicate the units, unless otherwise specified:

- Use SI units unless otherwise specified.
- There must be NO space between the answer and the units
- There must be between 1 and 3 digits to the left of the decimal point. *This means that you need enter 1kV rather than 1000V and 100mV rather than 0.1V*
- Write out 'ohm' to indicate Ω (use singular)
- Use 'u' to indicate micro.
- Only use pico(p), nano(n), micro(u), milli(m), kilo(k), mega(M), giga(G) and tera(T); do not use centi, deci and deca

If the answer is without units it has no meaning to write milli(m) or kilo(k). Simply write the number 0.1 or 1000

Results All Answers, Submitted Answers, Correct Answers, Feedback, Incorrectly Answered Questions

Question 1

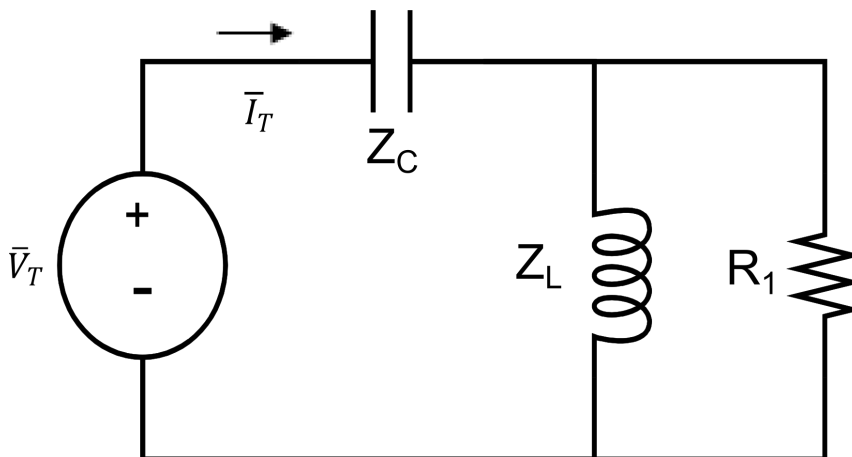
0.5 out of 2 points



In the figure below the capacitor is adjusted until the voltage phasor ($\bar{V}_T$) is in phase with the current phasor ($\bar{I}_T$). Specify the magnitude of the required impedance of the capacitor where this happens. (I.e. determine such that the phase of V and I differ by 0.)

Let $\mathbf{Z}_L = j7\ \Omega$ and $\mathbf{R}_1 = 19\ \Omega$

Do **not** include "j" in your answer, but include the unit



Selected Answer: ✗ 0.25ohm

Correct Answer: ✓ $6.163 \pm 5\%$ (ohm)

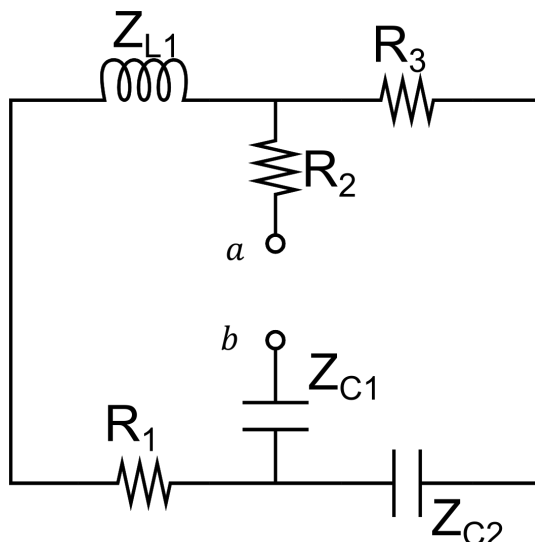
Question 2

0.5 out of 2 points



Determine the total impedance between terminals a and b in **rectangular form** and give B where **total impedance** = $\mathbf{A} + j\mathbf{B}$.

Take $\mathbf{Z}_{L1} = j28\ \Omega$, $\mathbf{Z}_{C1} = -j19\ \Omega$, $\mathbf{Z}_{C2} = -j26\ \Omega$, $\mathbf{R}_1 = 14\ \Omega$, $\mathbf{R}_2 = 1\ \Omega$ and $\mathbf{R}_3 = 16\ \Omega$.



Selected Answer: ✗ -56.7ohm

Correct Answer:  $-18.319 \pm 5\%$ (Ohm)

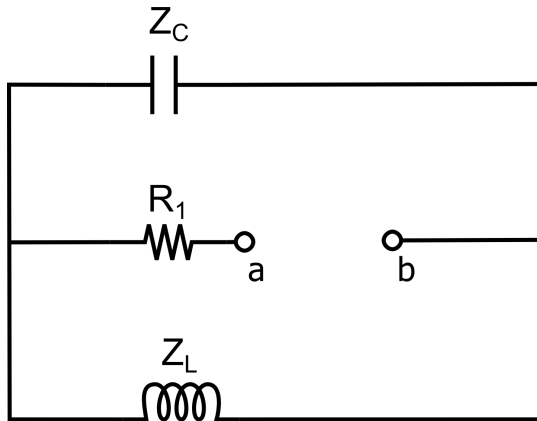
Question 3

2 out of 2 points



Determine the total impedance between terminals a and b . Solve in **rectangular form** ($A + jB$) and give A as your answer:

Let $Z_C = -j33 \Omega$, $R_1 = 54 \Omega$ and $Z_L = j22 \Omega$.



Selected Answer:  54ohm

Correct Answer:  $54 \pm 1\%$ (ohm)

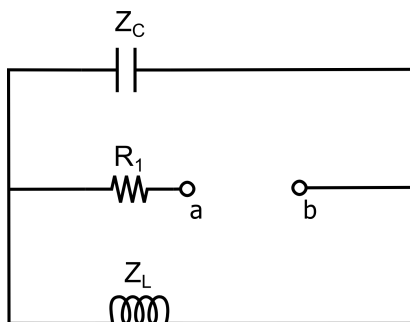
Question 4

0.5 out of 2 points



Determine the total impedance between terminals a and b . Solve in **rectangular form** ($A + jB$) and give B as your answer:

Let $Z_C = -j45$, $R_1 = 70$ and $Z_L = j45$



Selected Answer:  0ohm

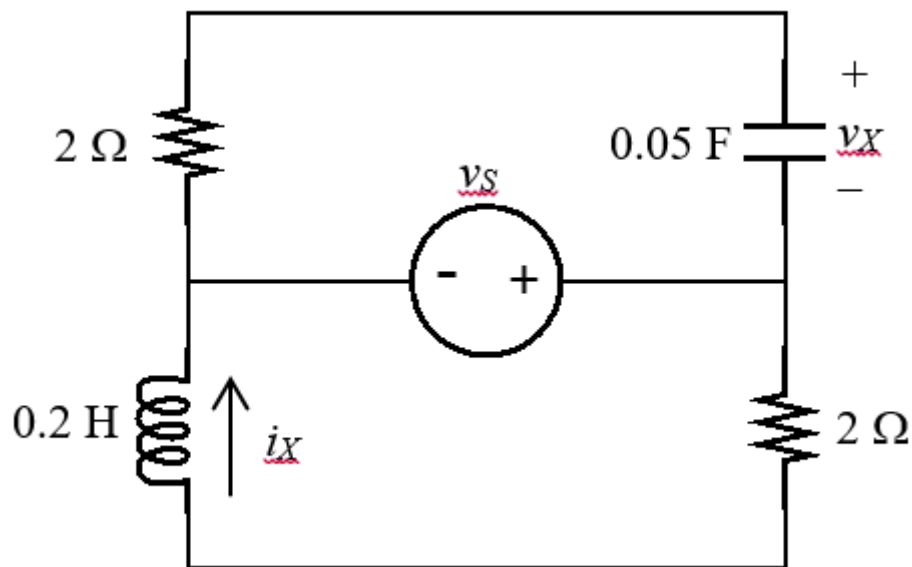
Correct Answer:  $-2,025,000,000.00 \pm 5\%$ (ohm)

Question 5

2 out of 2 points



Determine $V_x(t)$ given $V_s = 7\cos(10t + 45^\circ)$



The final answer must be in the following form: $\text{Acos}(10t + 180^\circ)$
Give the magnitude of A

Selected Answer: ☒ 4.95V

Correct Answer: ☒ $4.95 \pm 5\%$ (V)

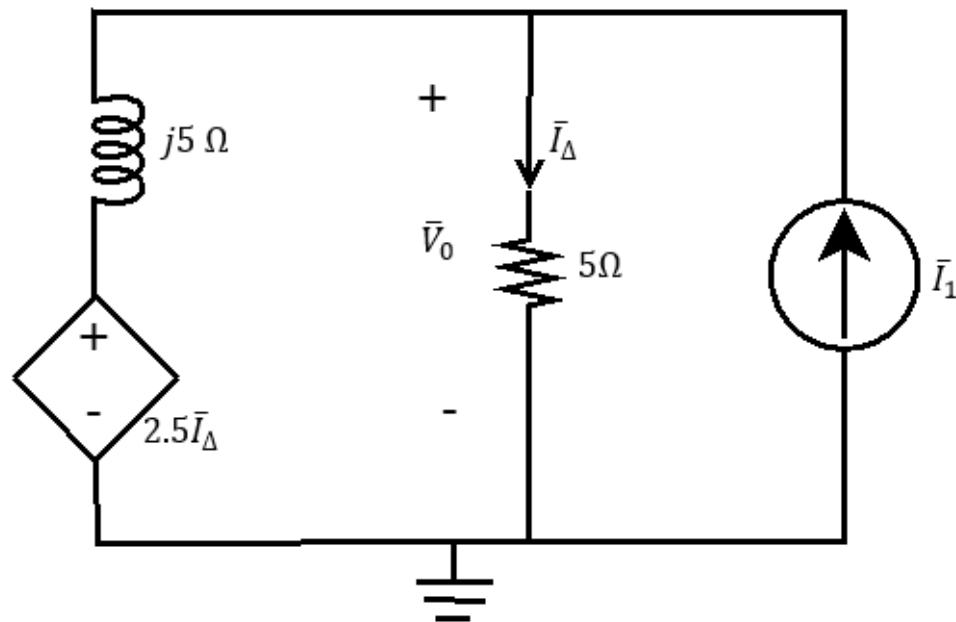
Question 6

0.2 out of 2 points



Determine $\bar{V}_0$ if $\bar{I}_1 = 65 \angle 0^\circ$. Write the answer in polar form ($a \angle b$).

Provide a.



Selected Answer: ☒ 58.1V

Correct Answer: ☒ $290.69 \pm 5\%$ (V)

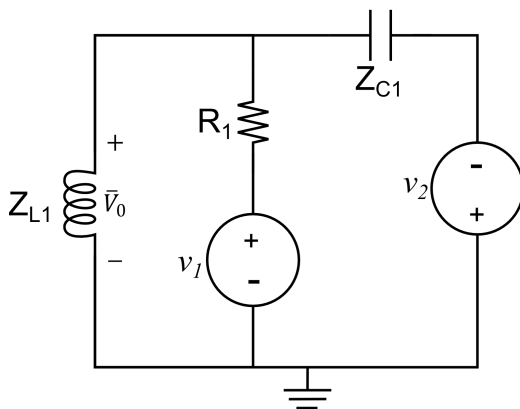
Question 7

2 out of 2 points



Provide the real part of V_0

Let $V_1 = j20$ V, $V_2 = 42$ V, $R_1 = 6$ Ω , $Z_{L1} = j14$ Ω and $Z_{C1} = -j5$ Ω .



Selected Answer: ☒ -14.7V

Correct Answer: ☒ -14.70 $\pm$ 5% (V)

Question 8

2 out of 2 points

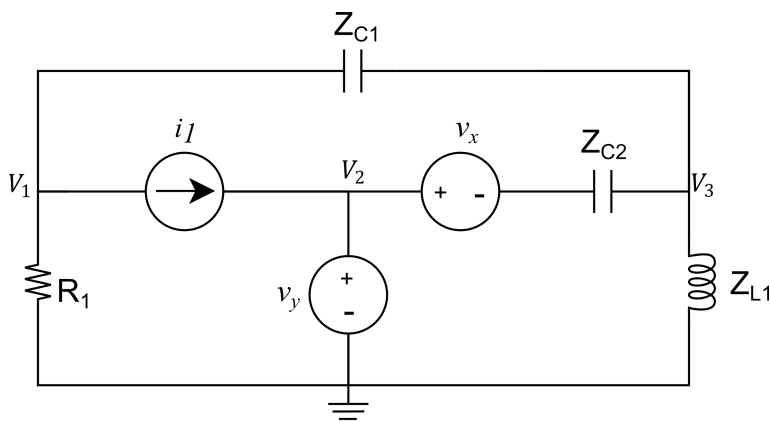


Use nodal analysis to set up an equation of the following form:

$$(a_1 + jb_1)V_1 + (a_2 + jb_2)V_3 = -24$$

Return a_2 as your answer.

Let $R_1 = 6$ Ω , $i_1 = 2$ A, $V_y = 3$ V, $V_x = 10$ V, $Z_{C1} = -j4$ Ω , $Z_{C2} = -j12$ Ω and $Z_{L1} = j1$ Ω .



Selected Answer: ☒ 0

Correct Answer: ☒ 0 $\pm$ 1%

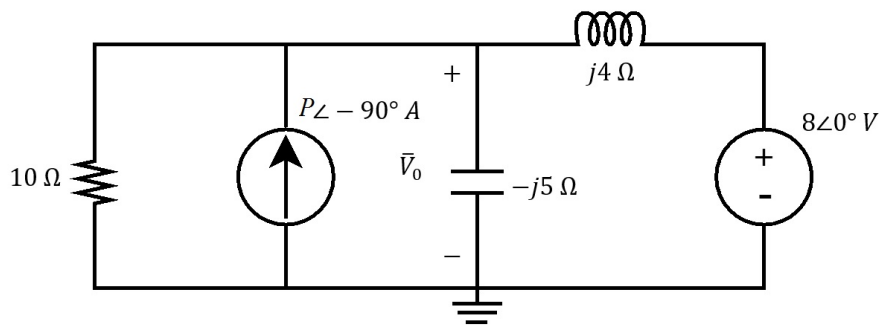
Question 9

2 out of 2 points



If $P = 9.8$ determine V_0 in rectangular format $A + jB$.

Give only the real component A .



Selected Answer: ☒ 47.2V

Correct Answer: ☒ 47.20 ± 2% (V)

Question 10

0 out of 2 points

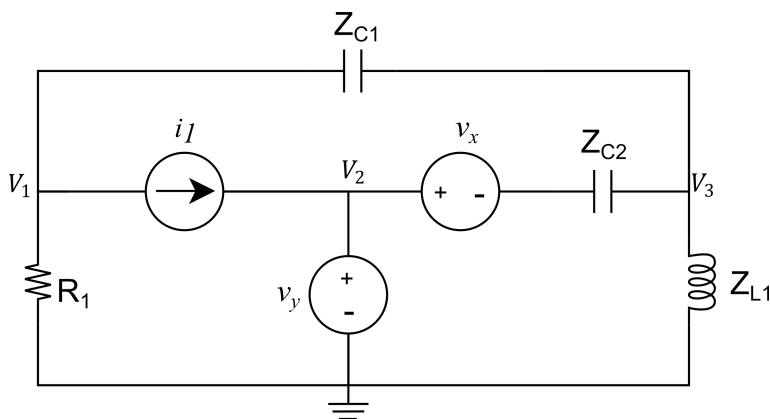


Use nodal analysis to set up an equation of the following form:

$$(a_1 + jb_1)V_1 + (a_2 + jb_2)V_3 = -24$$

Return a_1 as your answer.

Let $R_1 = 6 \Omega$, $i_1 = 5 \text{ A}$, $V_y = 7 \text{ V}$, $V_x = 16 \text{ V}$, $Z_{C1} = -j4 \Omega$, $Z_{C2} = -j22 \Omega$ and $Z_{L1} = j1 \Omega$.



Selected Answer: ☒ 2

Correct Answer: ☒ 0.80 ± 1%

Monday, October 23, 2017 5:12:58 PM SAST

← OK